Postural control and balance



Prof Isaac Sorinola

Learning Outcomes

By the end of this lecture, students should be able to:

- Understand and define basic balance terminologies (e.g., Centre of gravity, limits of stability)
- Explain the factors which define limits of stability
- Explain the differences between static and dynamic balance
- List the systems involved in balance
- Explain the basic control of human balance and factors that control it
- Describe the role of the vestibular apparatus, visual and proprioceptive systems, and the integration of these systems for balance control
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Balance: What is it?

Balance or Postural stability is the ability to maintain the body in equilibrium.

A body is in equilibrium either when it is at rest (static equilibrium) or when it is in steady-state motion (dynamic equilibrium)

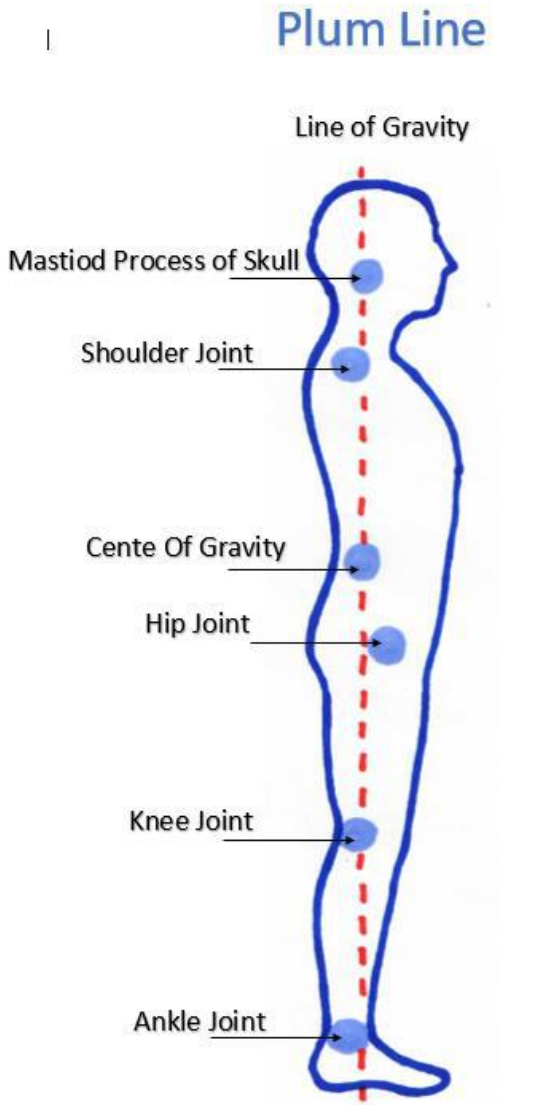
An object is considered stable or in equilibrium when the centre of gravity is over its base of support

Basic Balance Concepts: Line of Gravity

- The line of force passing vertically downwards through the Centre of Gravity (COG)

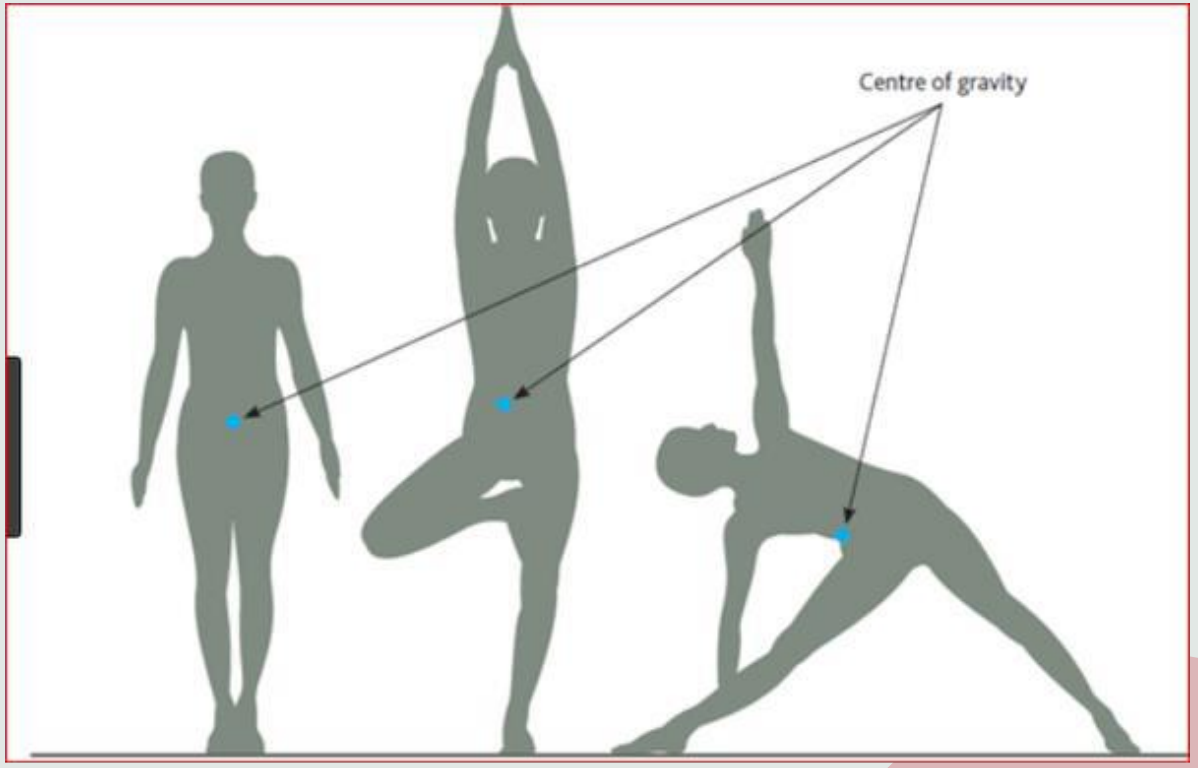
- Line of gravity falls from the vertex, through COG, in front of the ankles and within the base of support





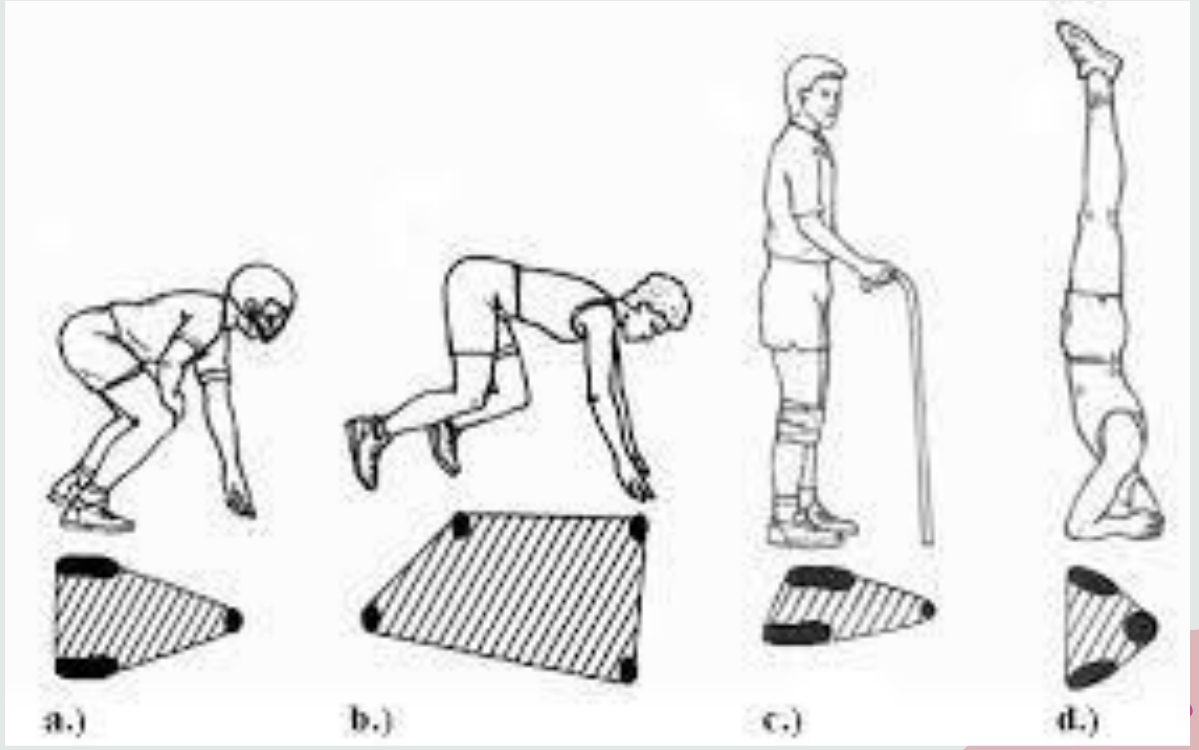
Basic Balance Concepts: Centre of Gravity (COG)

- The point where all the weight of the body is concentrated
- COG for human body in anatomical position: lower abdominal area + slightly forward of ankle joints.
- Segmentally each body area contains its own COG
- The COG moves with the position of the body



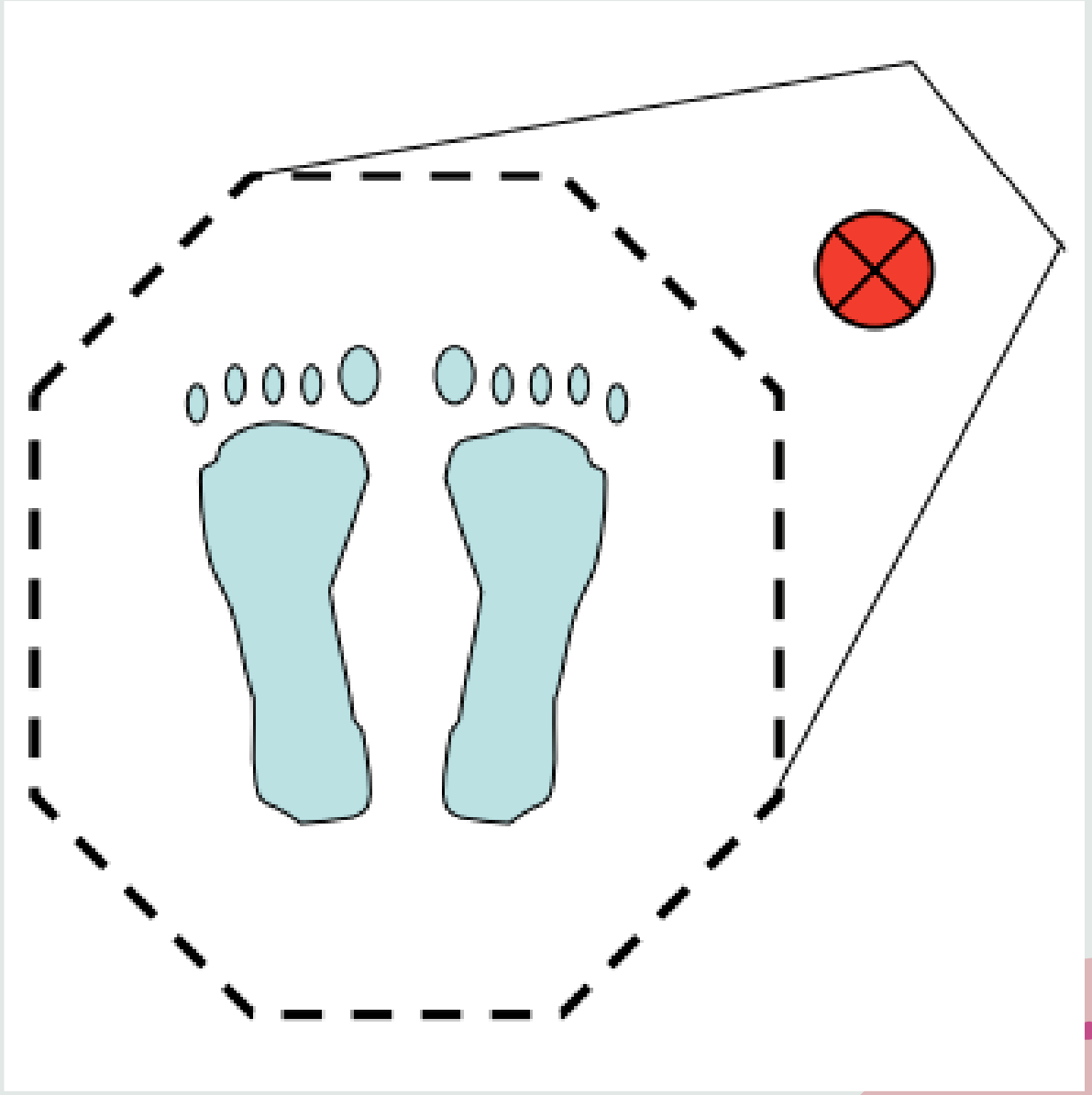
Basic Balance Concepts: Base of Support

Where body and support surface are in contact



Basic Balance Components: Limits of Stability

- Are the boundaries in which body maintains position without altering base of support
- When the limits of stability are exceeded, the individual must step, stumble, or grasp an external object to regain equilibrium. The alternative is a fall.
- ** “Perceived” Limits of stability may differ from actual limits of stability



Normal and Abnormal Limits of Stability



What defines the limits of stability

1. **Position of line of gravity within the base of support: Maximal stability when line of gravity passes in the centre of the base of support**

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2. **Area of BOS: Larger base = more stable**



What defines the limits of stability

1. Position of line of gravity within the base of support: Maximal stability when line of gravity passes in the centre of the base of support
2. Area of BOS: Larger base = more stable
3. **Height of COG: Lower COG = Greater stability**



What defines the limits of stability

1. Position of line of gravity within the base of support: Maximal stability when line of gravity passes in the centre of the base of support
2. Area of BOS: Larger base = more stable
3. Height of COG: Lower COG = Greater stability
4. **Body Mass: Greater mass = Greater Stability Why? Force = mass x acceleration**



What defines the limits of stability

1. Position of line of gravity within the base of support: Maximal stability when line of gravity passes in the centre of the base of support
2. Area of BOS: Larger base = more stable
3. Height of COG: Lower COG = Greater stability
4. Body Mass: Greater mass = Greater Stability Why? Force = mass x acceleration
5. **Velocity of COG: Slow sway = Greater stability**
    - **Slow sway oscillations (front to back or side to side and back again last 2 to 3s) gravity is only significant destabilising force that must be overcome.**
    - **Rapid sway oscillations (1s or less) momentum of body is additional destabilising force**





Balance Impairments

From clinical perspective a patient may have impaired balance because:

1. The position of the COG relative to the base of support is not accurately sensed
2. The automatic movements required to bring the COG to a balanced position are not timely or effectively coordinated
3. Musculoskeletal injury/ Insufficient muscle strength/damage to individual sensory systems
4. Combination of the above.

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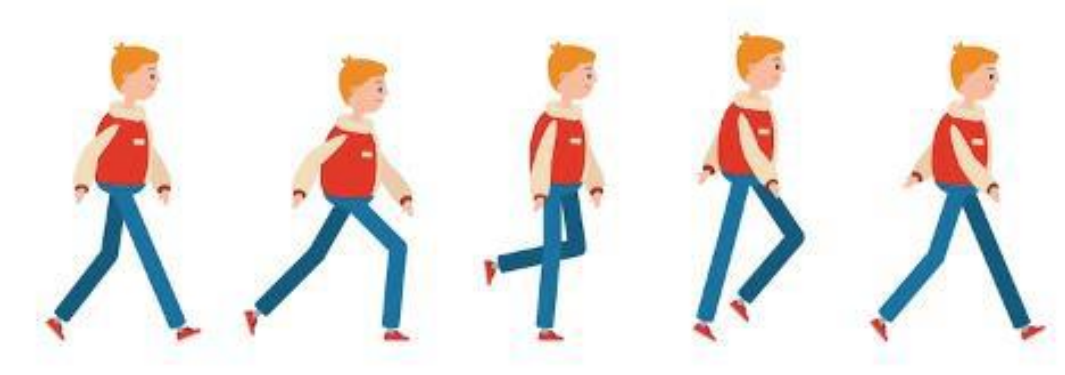
Balance: Static vs Dynamic

Static balance

- Ability to hold the body in a specific position and posture e.g., lying, sitting, standing postures
- Body remains stationary and COG is over base of support
- Co-ordinated muscle contractions are required to ensure stable equilibrium.



Dynamic Balance - Gait



- Many dynamic movements including walking and running are UNSTABLE
- Walking and running essentially consist of a series of interrupted falls
- Before there has been time to fall base of support is moved so equilibrium is restored

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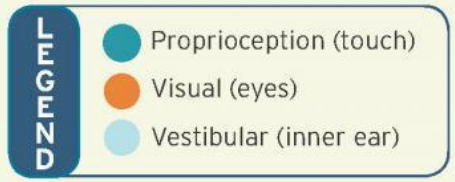
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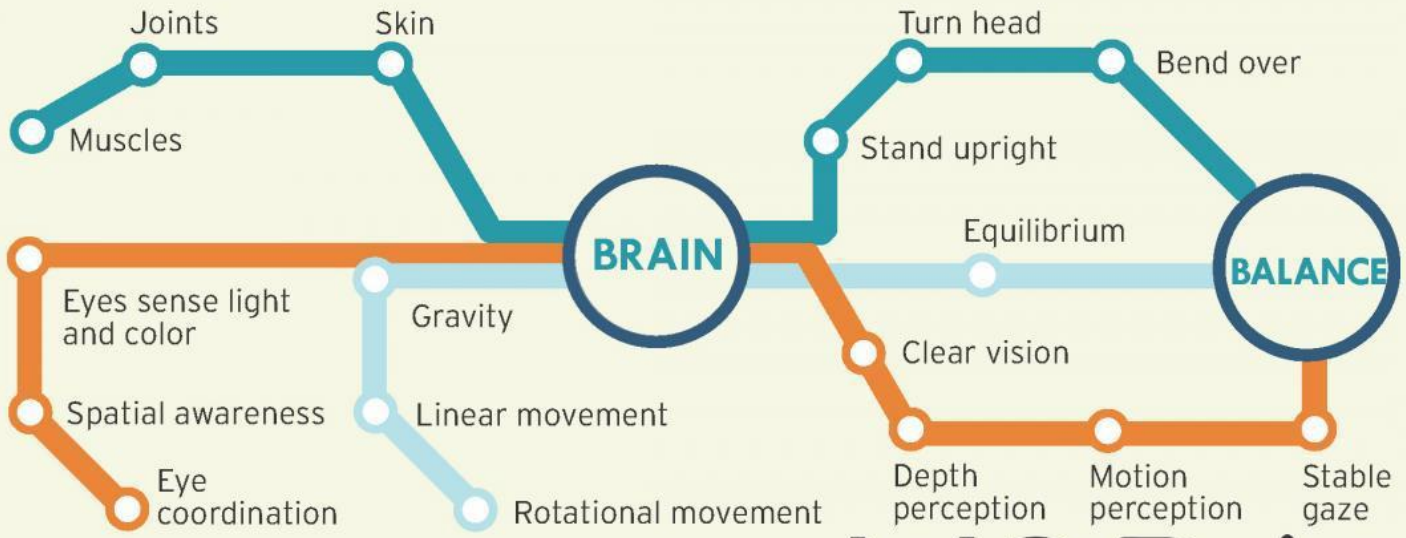
Balance control



THE ROAD TO BALANCE

The brain combines inputs from the vestibular system, eyes and muscles (proprioception) to give us good balance.





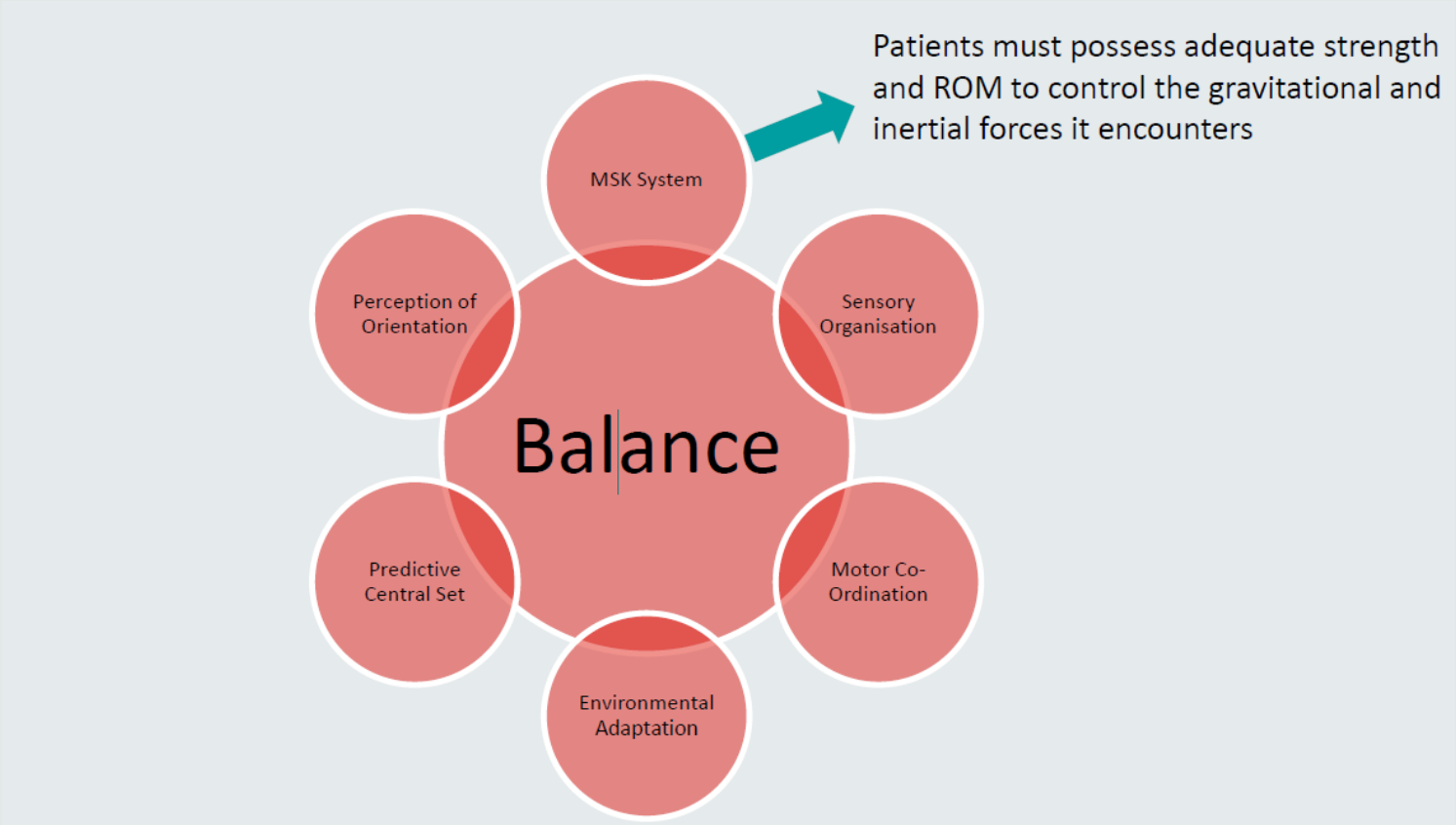
HAVING TROUBLE FINDING BALANCE?

Visit vestibular.org

VEDA

LIFE REBALANCED

Balance System – How does it work?



Balance System – How does it work?



Organisation and interpretation of the raw information we receive from the three sensory systems.

Balance System – How does it work?

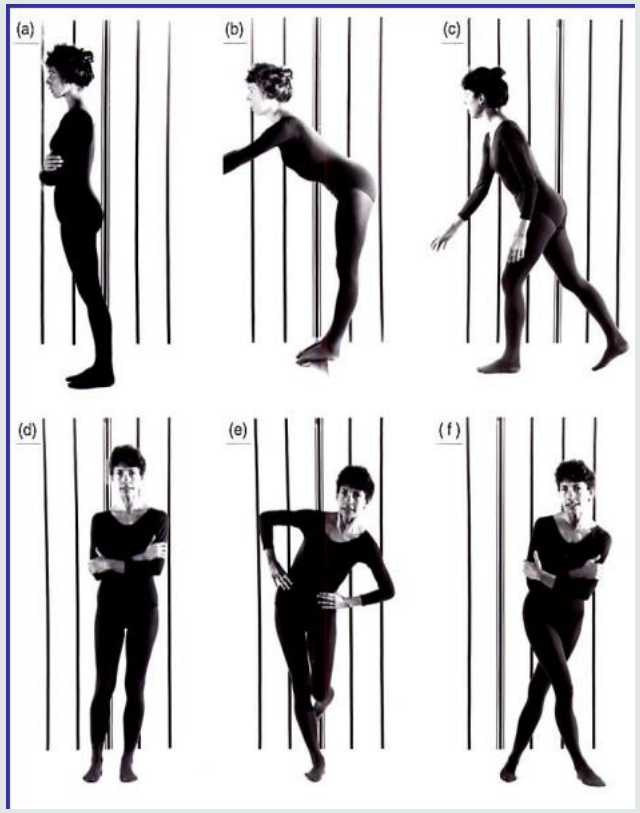


Temporal and spatial sequence of muscle activation. Balance behaviour, while complex, consists of "a limited repertoire of central motor programmes" called muscle synergy

Postural movement strategies during perturbed stance

ankle hip stepping

Ankle strategy	**Hip strategy**	**Stepping strategy**
Forward sway: Gastrocnemius Hamstrings Paraspinal muscles	Forward sway Abdominal muscles Quadriceps	Forward perturbation Forward step
Backward sway Tibialis anterior Quadriceps Abdominal muscles	Backward sway Paraspinal muscles Hamstrings	Backward sway Backward step



Anterior/posterior direction

Lateral direction

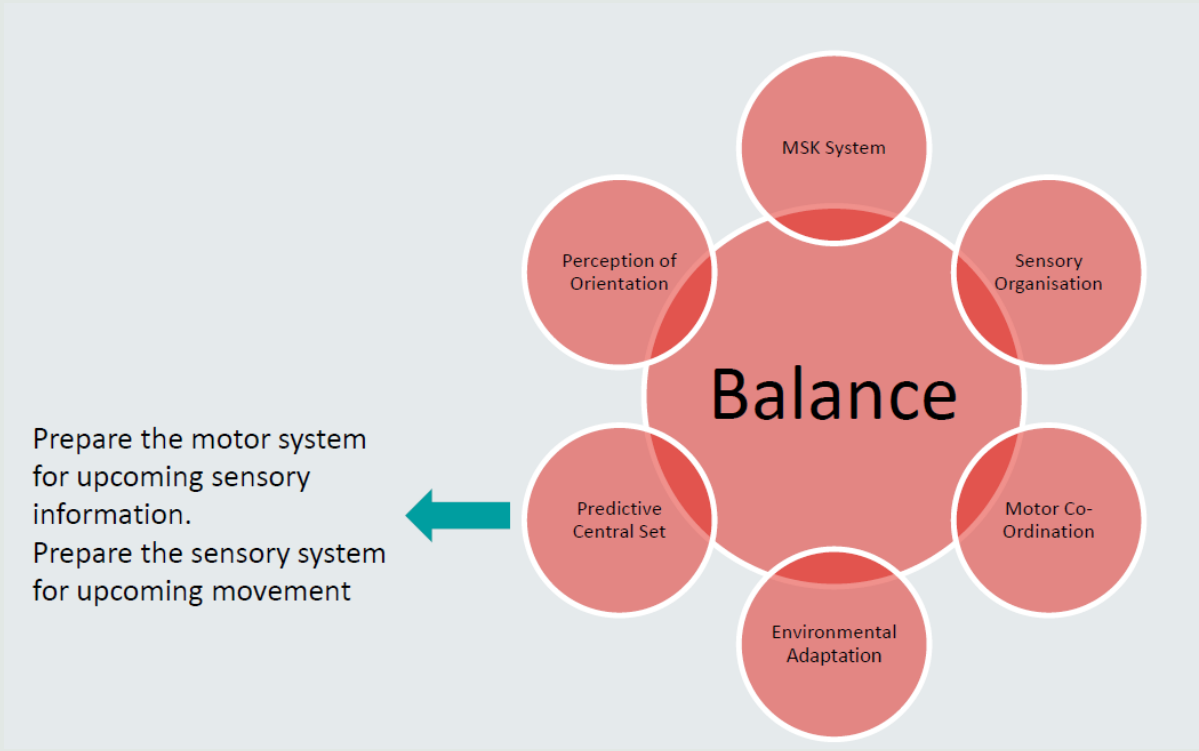
Hip abductors/ adductors
Ankle movements

Balance System – How does it work?



The motor strategy a person uses depends in part on the type of surface, or on whether they are sitting or standing.

Balance System – How does it work?

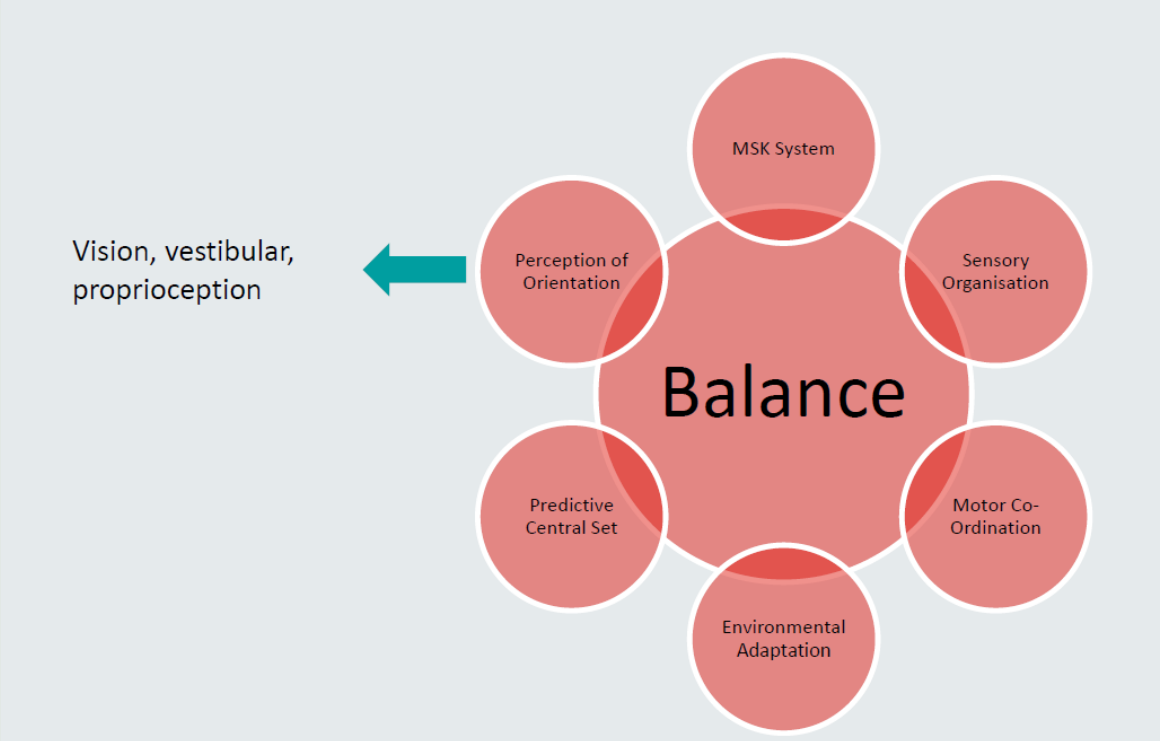


Example of anticipatory postural adjustments

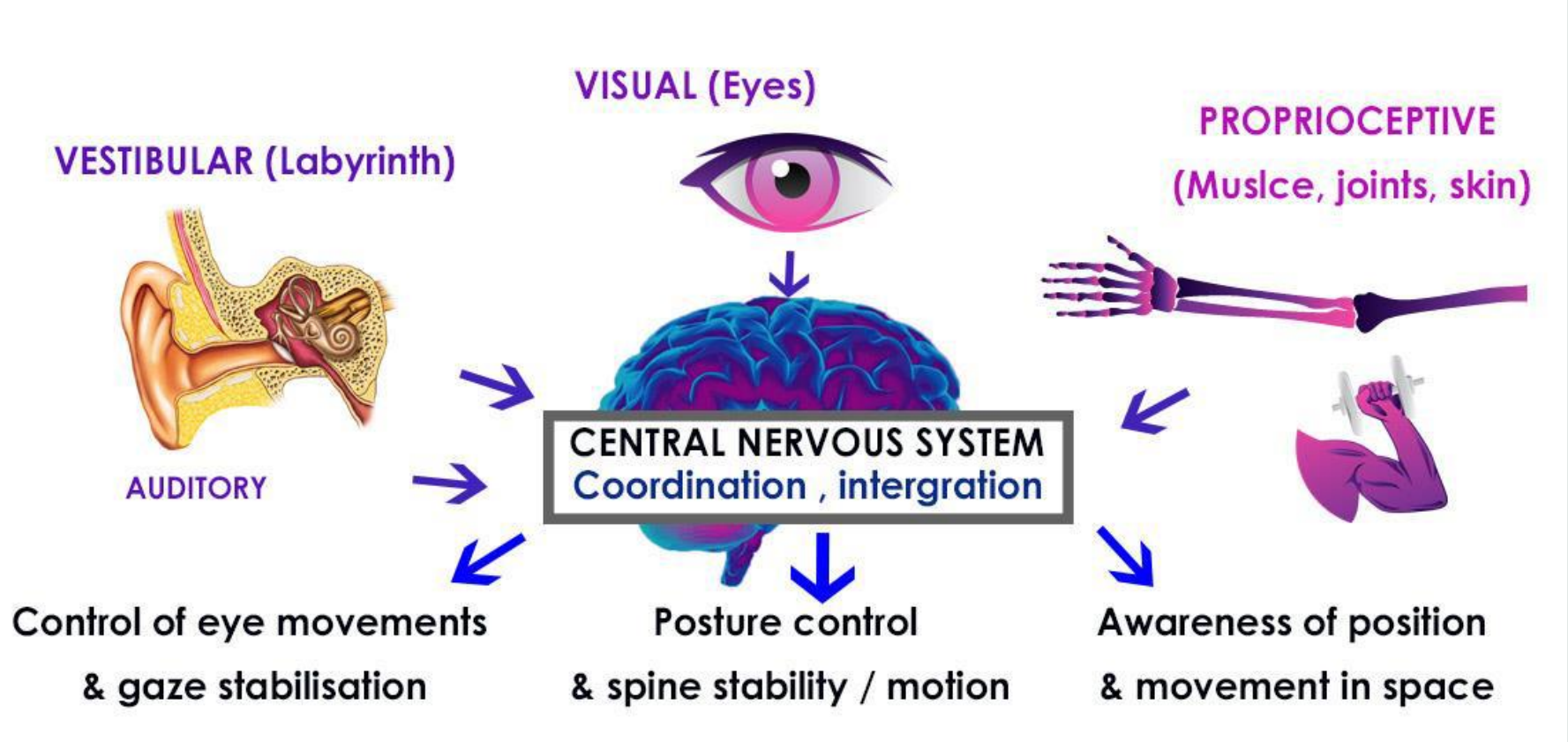
When lifting a heavy bag, anticipatory postural adjustments accompany a potentially destabilising limb movement



Balance System – How does it work?



The balance system: Perception of orientation



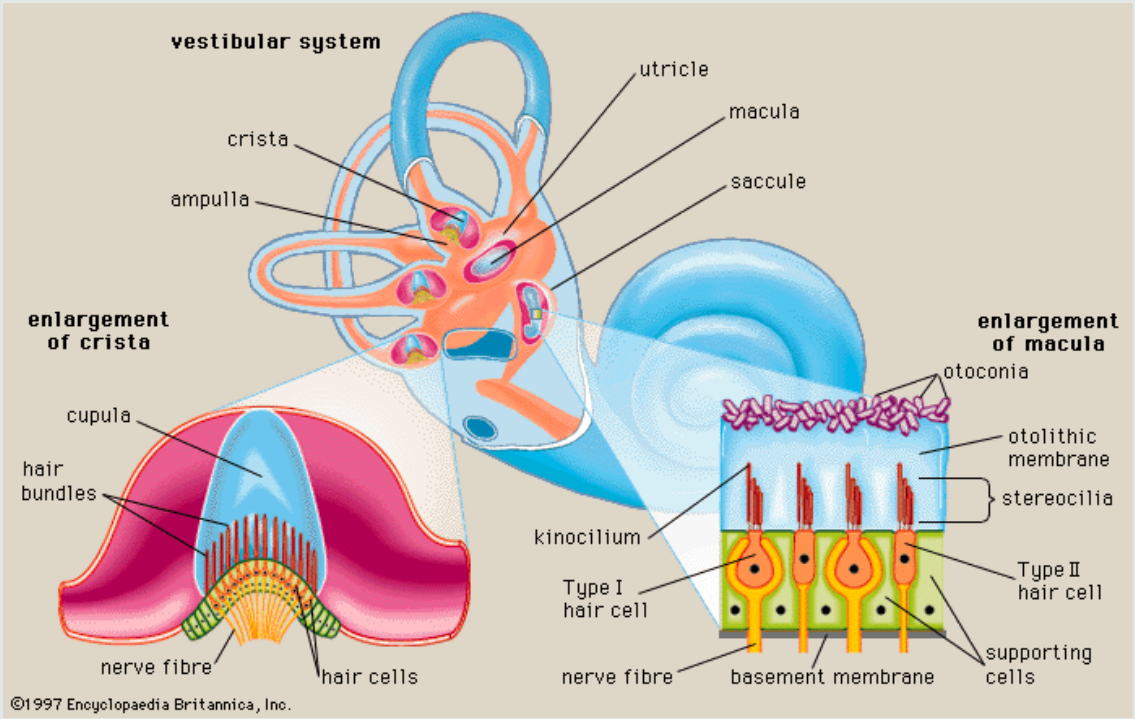
Perception of orientation: Visual inputs

- Used to sense and judge the world around us
- Provides information to both the vestibular and somatosensory systems

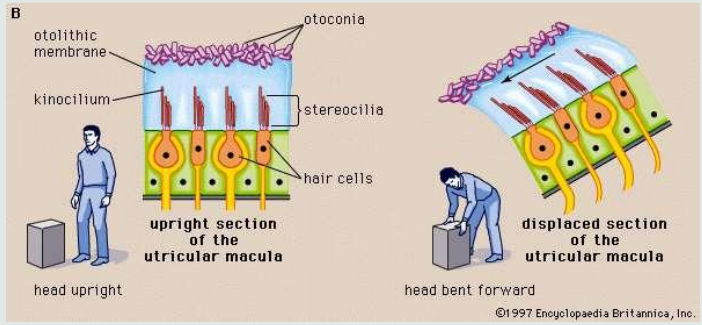
Feedback needed to maintain balance:

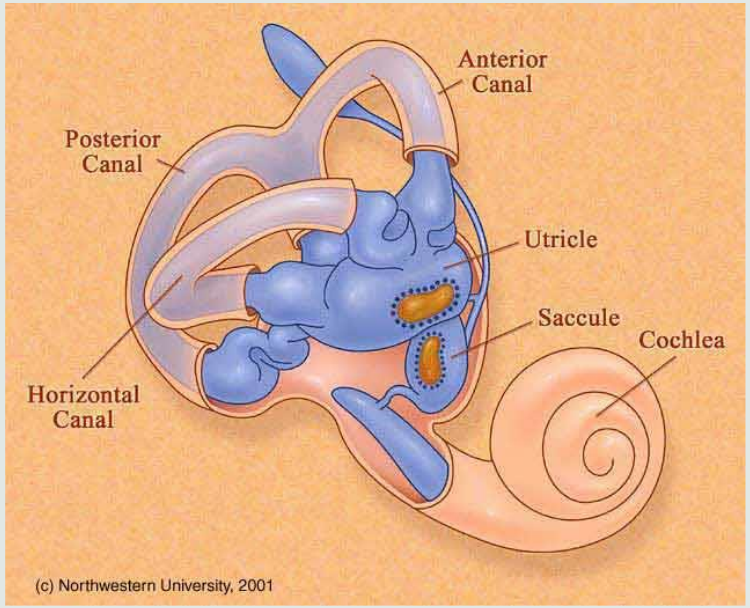
- The position and movement of head with respect to surrounding objects - reference for verticality
- The direction of vertical
- Depth and motion perception

Vestibular system

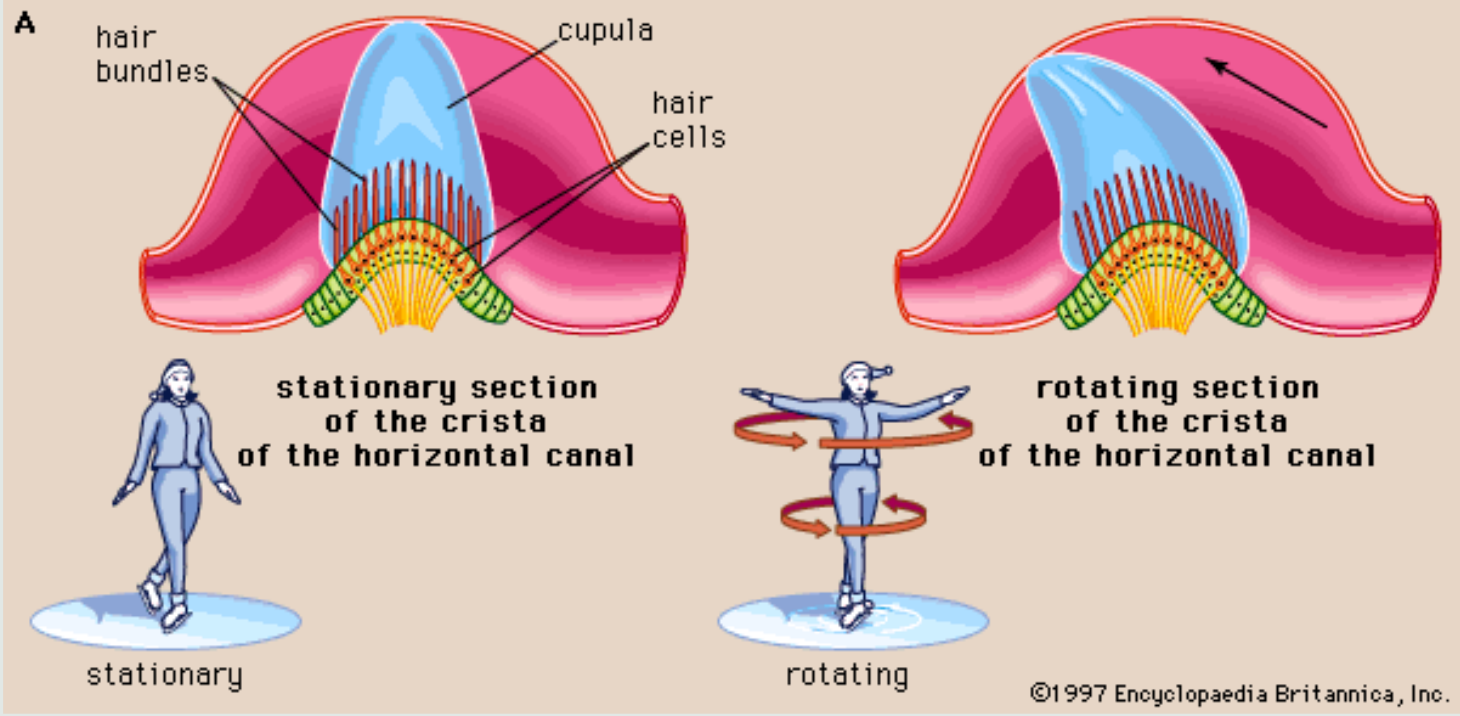


Vestibular System –Linear Acceleration

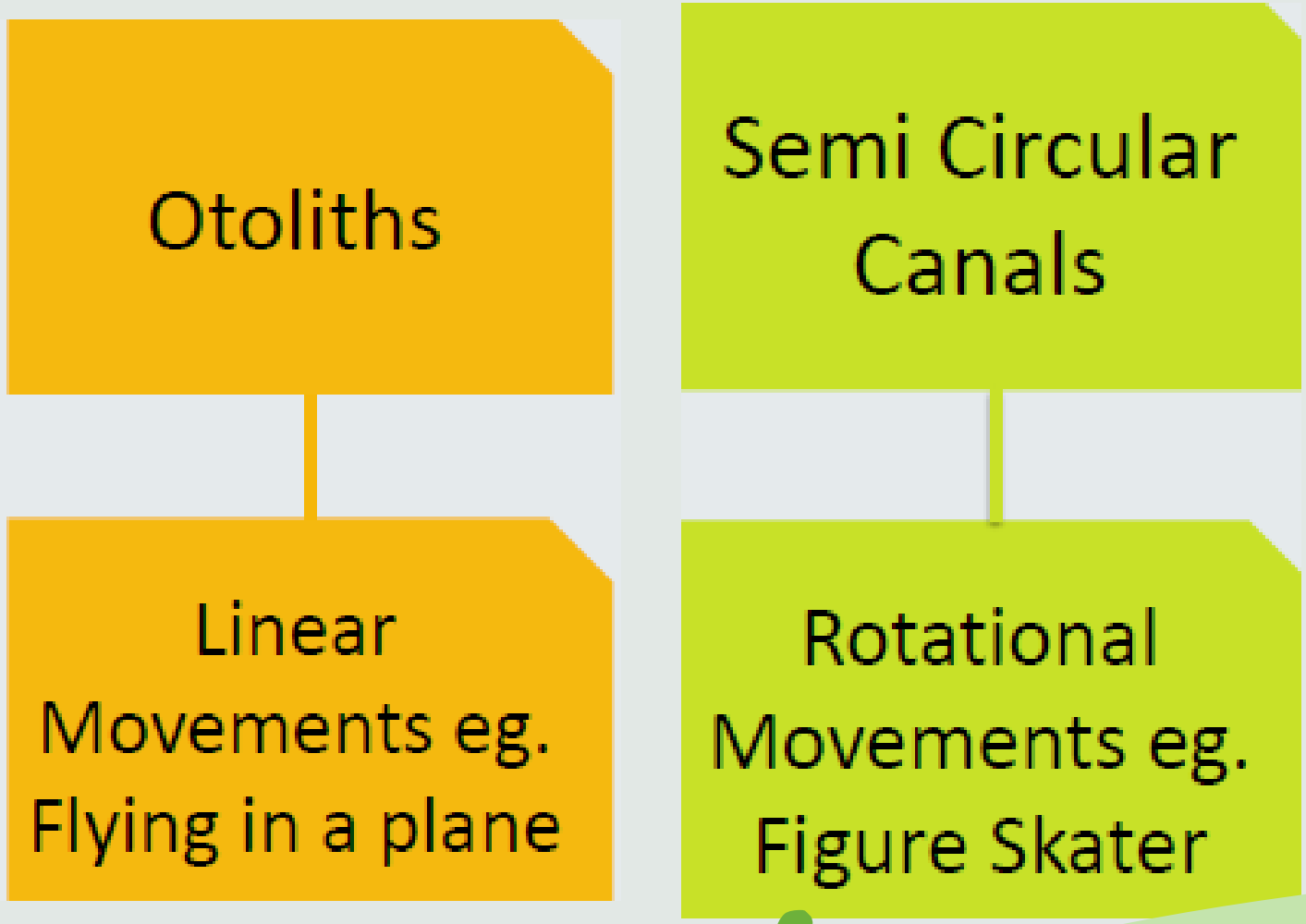




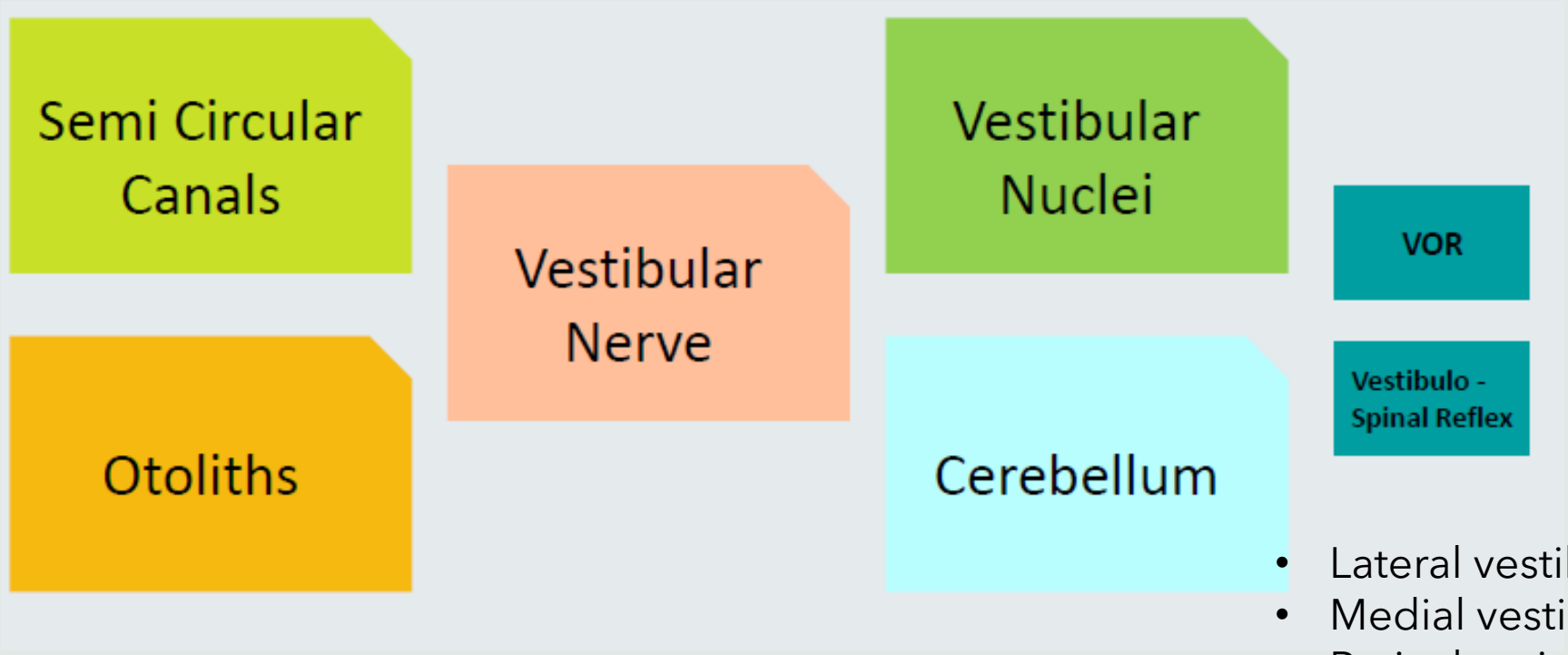
Vestibular System – Angular Acceleration



Vestibular System

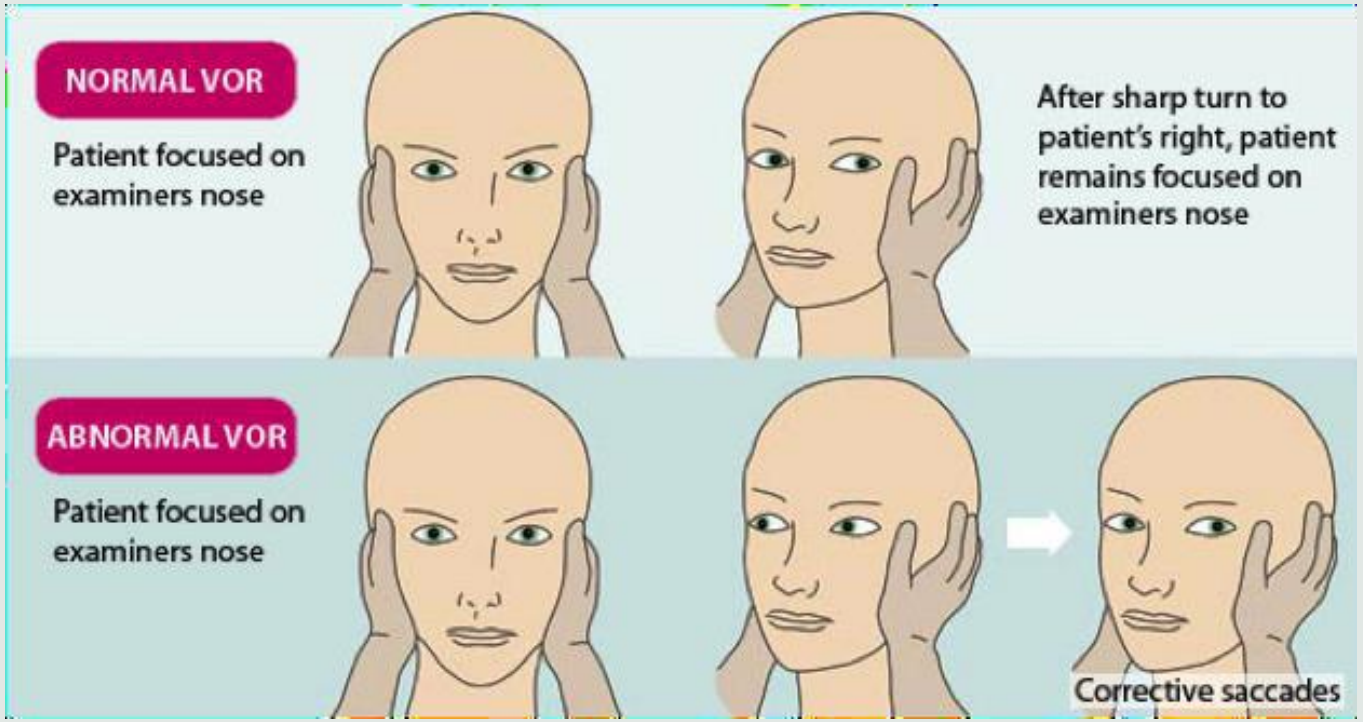


Vestibular System



- Lateral vestibulospinal tract
- Medial vestibulospinal tract
- Reticulospinal tract

Vestibulo-ocular Reflex (VOR)



Testing Vestibulo-ocular Reflex



Vestibular system

Role of Vestibular System:

- Determines position and motion of head in space
- Gaze stability (VOR)
- Maintain head and posture stability

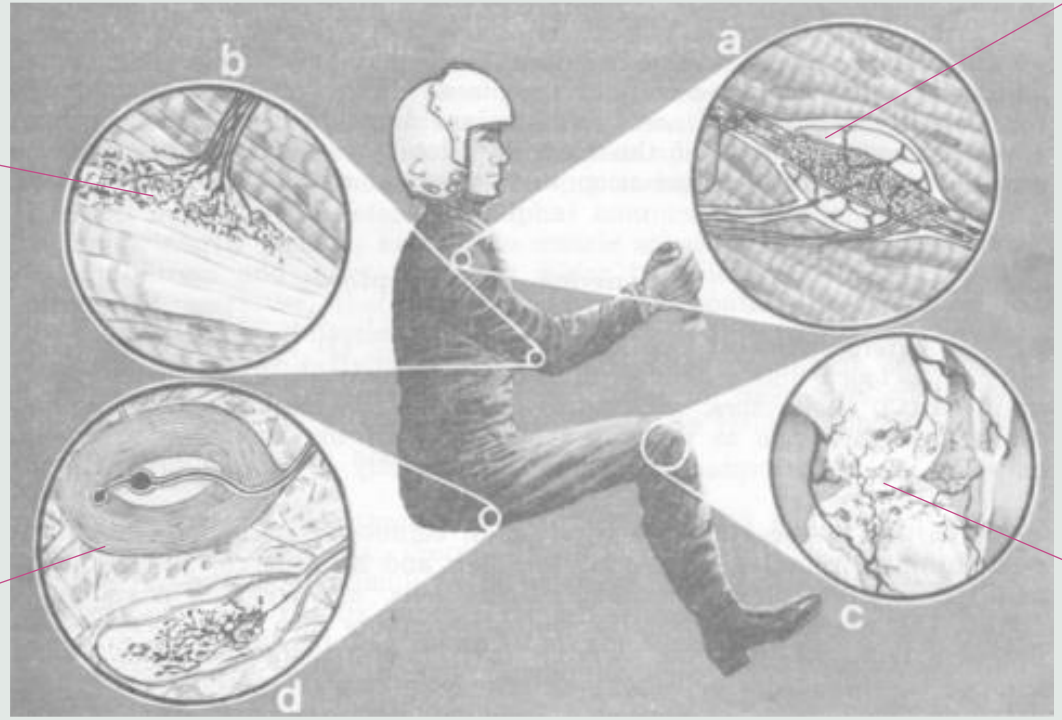
Feedback needed to maintain balance:

- Velocity of head motion
- Head position with respect to gravity and other inertial forces (e.g. moving vehicles)
- Orientation to vertical

Somatosensory system

Muscle spindle

Golgi tendon organ



Mechanoreceptors: Pacinian and Ruffini corpuscles

Joint receptors

Somatosensory system

The position of the body with respect to its support surface

The position of body segments with respect to each other



Balance System – How does it work?



Adapting sensory organisation to changing contexts

Inter-modal theory

All three senses always contributes equally

The inputs increases specificity and never conflicts

It is the relationship between the inputs that allows sensory orientation to emerge

Sensory weighting hypothesis

CNS modifies the weight or importance of a sensory input

CNS resolves conflicts between the sensory inputs

The gain on an input depend on its accuracy

There is more research evidence to support this

Cognitive and balance control

- Role of **attention** and **executive functions** in balance control
- Postural sway and functional balance impairment > in people with cognitive impairment.
- Greater cognitive impairment is associated with poorer balance control

Cognitive and balance control

- Specific cognitive domains are associated with fear of falling in older adults and people with neurological conditions

- Slower gait speed, shorter stride length, increased stride width, and prolonged double limb support time have been found to be associated with fear of falling in older adults and people with neurological conditions

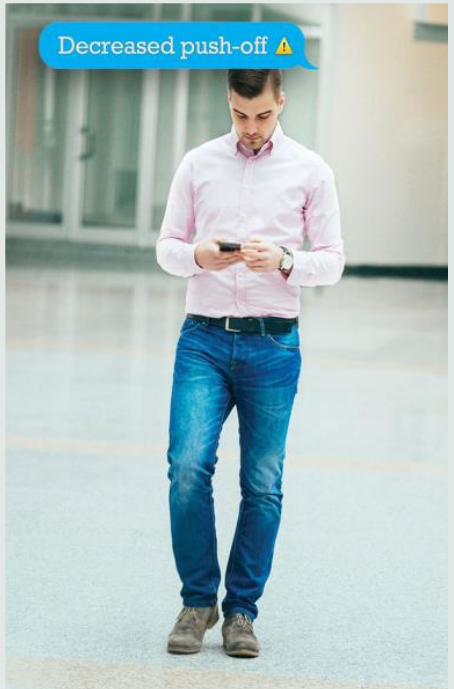
- Growing body of literature investigating the effect of cognitive and/or balance training on cognitive function domains and postural and gait parameters with varying results

What happens when walking and texting: dual task activity



Absence of arm swing

Decreased gait speed





Increased co-contraction of the ankle antagonist muscles

= increased risk of injury and falls

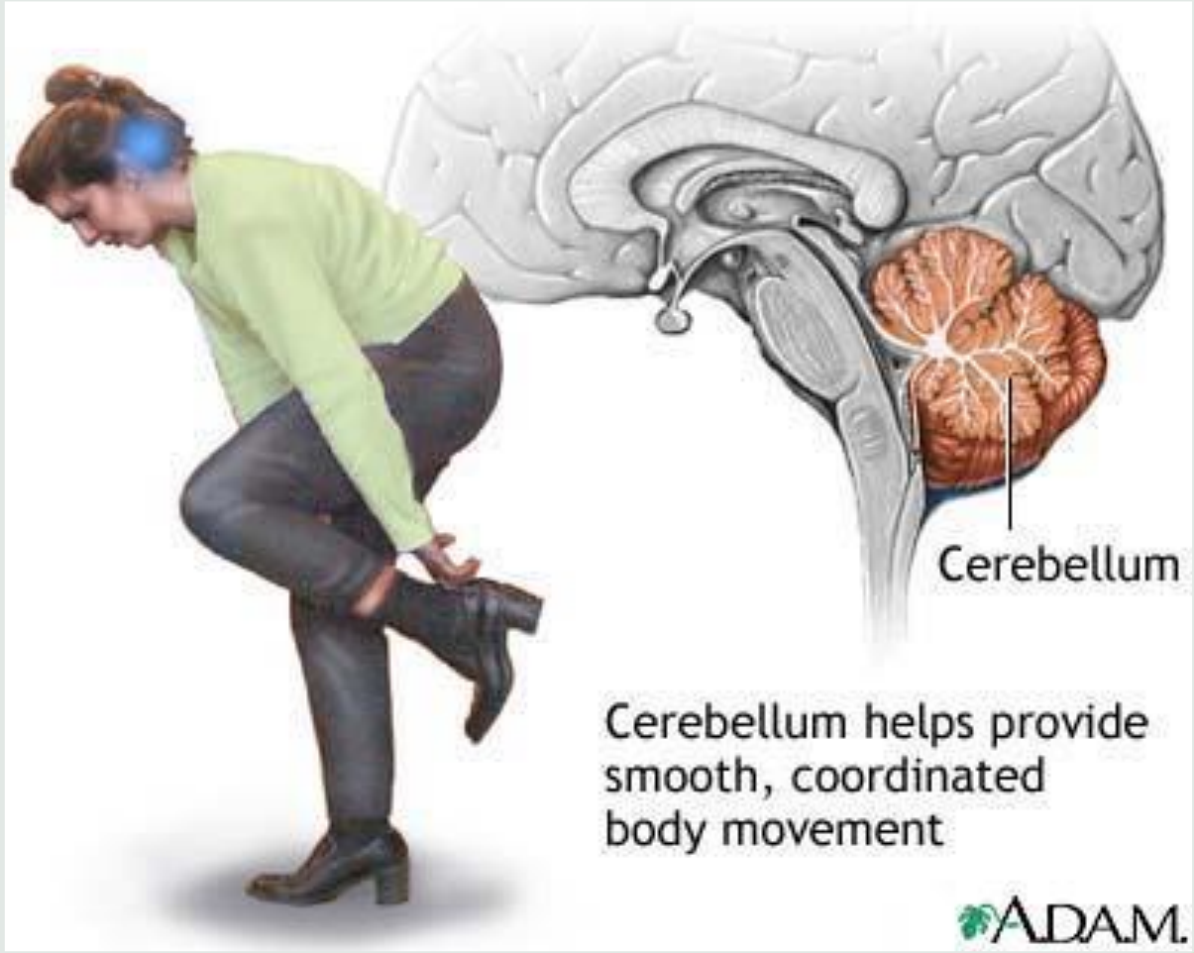
Cerebellum

Involved in co-ordination of movements

MONITORS

COMPARES

CORRECTS



Summary

Postural control is a complex motor skill based on interaction of dynamic sensorimotor processes.

Balance function depends on strategies that individuals use to accomplish stability according to environmental and task demand.

Balance impairments result from damage to either sensory or motor components of balance system